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Who came up with E=m c^2? /

Most people don't know this but an Italian, Olinto De Pretto, derived the E/m=c^2 relation with a simplistic argument back in 1903, with priority over Poincare. Also, Einstein never proved the relation in general [3]. So he wasn't the first to come up with it, and he wasn't skilled enough to prove it.

Olinto De Pretto argument couldn't be simpler (even if it was wrong in a kinetic sense): he used the vis viva (E=mv^2) and argued that matter is made of light so inside of it somewhere, v=c, so E=mc^2. We know the kinetic energy (non-relativistic) is E=mv^2/2 today so the vis viva of Leibniz, though correct in its conservation context, doesn't work when you consider dynamics. So his argument cannot stand on its own, even though he got the correct result.

Besides, there was several priorities over Pretto. The ingenious Russian Nikolay Umov (of the Umov-Poynting vector fame), had already constrained E/mc^2 to between 0.5 and 1. He used the aether concept to do this, all the way back in 1873. This is the first known attempt at finding the energy-mass equivalence relationship, a concept grounded in the aether concept. Today his proof isn't accepted as having any priority over modern incarnations because academia hates the ancient concept of the aether, also known as the quintessence, or the fifth element.

Later, in 1875, the English Samuel Tolver Preston, gave another aether argument in his book Physics of the Ether, where he derived E=mc^2 using a heuristic argument. He also was the first (as far as I remember) to propose the theoretical conversion of particles into light which was a premonition of what would become the principle operating procedure of the nuclear bomb [1]:

"[A] quantity of matter representing a total mass of only one grain, and possessing the normal velocity of the ether particles (that of a wave of light), encloses a store of energy... not less than that possessed by a mass of forty thousand tons moving at the speed of a cannon ball."

(And now you know why I looked up this stuff, lol)

Preston also considered light and matter interchangeable, so the equation worked in reverse (light has a mass-like effect).+

Meanwhile, in his 1905 paper, Einstein was clearly plagiarizing someone else's work by giving a circular argument for the same result. This was pointed out by countless people, including Max Planck, and really you can see it for yourself in his paper, he assumes that

which he claims to prove.

I actually think it was likely Poincaré whose work was his target. Poincaré 🇫🇷 gave a proof of the E/m ratio by integrating the total field energy assuming some radius R and dividing by the momentum in the Poynting vector for a small motion that creates a magnetic field, resulting in cancelation and the formula (4/3)mc^2 that others had derived before him. Then he corrects for necessarily internal pressure to keep his particle structurally sound giving us the formula we know and love. He used aether arguments (equating the forces derived from Maxwell's stress tensor to actual forces) in his proof which is an important point.

In the same paper Poincaré also explained the consequences of the M-M experiment's failure to detect drag as not surprising since we couldn't see any shifts in terrestrial optics, meaning that the medium of propagation (and yes there is a medium, analogous to a stiffness epsilon0 and density mu0) had to obey Lorentz invariance, which is actually quite fascinating when you dive deep into the consequences especially with rotation. By the way, this doesn't mean Poincaré's proof was sound*. Actually, E/m=c^2 is built into the aether concept if you identify mu0 with density.

Humble as ever, Poincaré called all of this the "postulate of relativity", citing all who worked on it before him like a decent person would. Later in 1906 he produced a work that contained the parts needed to put together full energy relation E^2=m^2c^4+p^2c^2, albeit in a scatter form of postulates. Our good friend Max Planck put it together in the same year, citing Lorentz and Einstein but sadly not Poincare.

Einstein on the other hand, claimed the aether was "superfluous", and for that reason alone, claimed that means it cannot exist. Which is of course nonsensical, but what better way to undermine the actual proof to promote your own circular argument is there? Consequences for humanity be damned. He later branded it the "special theory of relativity", promising one that unites gravity later.

When it was pointed out to him that his proof was circular, he tried many many times to produce a new one without using the same aether style argument as Poincare. He failed. Max von Laue 🇩🇪 later used the same stress-energy tensor proof as Poincaré but it was from another angle -- the radiation contributes extra momentum, so it must be subtracted out. Felix Klein supposedly proved it in a more rigorous manner but I've never looked at the proof closely [2]. Ultimately, speaking of density i.e. mass without a medium is nonsensically abstract, more mathematical than physical.

Ironically, Einstein's underhandedness would get the better of him later. When he was working on a framework for accelerating frames, which he called general relativity, Einstein began to reform. After being humbled by his inability to prove E/m=c^2 and the requirement for a variable speed of light in Jahrbuch für Radioaktivität (1907), he started to adopt the ether idea again. This all culminated in his 1920 lecture, titled "Ether and the Theory of Relativity", Einstein said:

"[Special relativity] does not compel us to deny the ether. We may assume the existence of an ether; only we must give up ascribing a definite state of motion to it"

He then went even further:

"According to the general theory of relativity, space is endowed with physical qualities; in this sense, therefore, there exists an ether. Space without ether is unthinkable; for in such space there not only would be no propagation of light, but also no possibility of existence for standards of space and time"

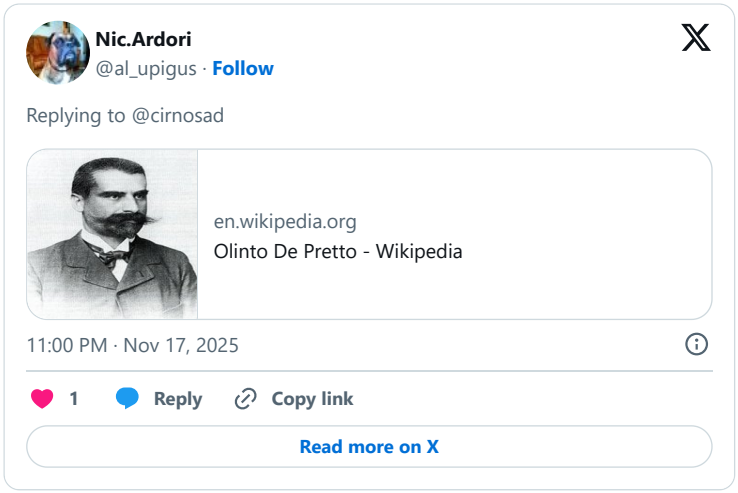
It's important to note that if suddenly the aether was fashionable again, then priority would go to those who had used it to derive E=mc^2 again.

... But by the time Einstein had realized his mistake, had already been too late. Earlier in 1915, Hilbert** 🇩🇪 and other messy differential geometry mathematicians join in, adopt his no-ether/constant speed of light framework, and necessarily bend space instead. It ended up being an awful mess that crippled the mind of astro-soyboys ever since, robbing humanity of the insight of relativistic plasma physics for generations to come. It also kept Einstein out of that line of research since he wasn't into the abstract differential geometry they were used.

Today, despite plasmas making up 99.9% of the visible universe around us, everyone focuses on masses. An ironic inversion of the focus on the mass-energy relation. Space is thought of as empty, when in fact it is full, yet somehow it can be bent and twisted around resulting in singularities (blackholes). And it's all thanks to one underhanded attempt at getting noticed.

/End

N.B I use m here but it's really m0. m is the corrected mass under gamma.



Un docente di storia della matematica sostiene che la scoperta del fisico tedesco

Il giallo della formula di Einstein

Lo scienziato passato alla storia per la "teoria della relatività" pubblicò i primi risultati nel 1905. Oggi si è scoperto che già un anno prima il vicentino Olinto De Pretto era giunto alle sue conclusioni

Che delusione, se fosse vero... Il mitico Albert Einstein, ritenuto il più grande genio della fisica moderna, vincitore del Premio Nobel nel 1921 per aver scoperto la "teoria della relatività", era forse un inguaribile ambizioso. Oggi si dice che, pur di raggiungere la gloria, aveva scopiazzato e fatto suoi gli studi di uno scienziato italiano poco noto, tale Olinto De Pretto, di Schio in provincia di Vicenza. Dello straordinario talento di De Pretto si è venuti a conoscenza solo ora. Se si legge il libro del professor Umberto Bartocci, "Albert Einstein e Olinto De Pretto. La vera storia della formula più famosa del mondo", i dubbi sull'aver "scopiazzato" la formula da parte di Einstein si rafforzano. La formula è la celeberrima E=mc2, alla base della teoria della relatività, e Bartocci, docente di storia della matematica all'Università di Perugia, pro-

celli, sua pronipote nonché titolare dell'archivio di famiglia.
La signora Bonicelli, ex insegnante di filosofia, così ci descrive l'antenato: «Olinto De Pretto era un uomo riservato che trascorse gran parte della vita svolgendo l'attività di direttore amministrativo presso la fonderia del fratello Silvio e dedicandosi allo studio delle più svariate materie: l'agraria, la geologia, l'astronomia, la fisica... Il suo carattere, per natura chiuso, si apriva con estrema disinvoltura quando era in compagnia degli amici, conosciuti nei circoli culturali oppure durante le escursioni in montagna, l'unico svago che Olinto si concedeva. Nelle vecchie lettere e nei documenti che possiedo viene definito come un vero e proprio gentiluomo: cordiale, educato, umile. Non si è mai sposato, ma aveva una forte concezione della famiglia. Lo si capisce dallo splendido rapporto che lo legava ai cinque fratelli e alle

FOTO DE BELLIS

È sua la teoria che ha cambiato il mondo?
Nato a Schio, nel Vicentino, Olinto De Pretto (nella foto) avrebbe inventato la formula che ha portato alla costruzione della atomica.

Scoprì la legge più importante del- nello scritto "Ipotesi dell'Etere

[1] https://archive.org/stream/physicsether01presg00g/physicsether01presg00g_djvu.txt#:~:text=a%20total%20mass%20of%20only,contains%20an%20energy%20not%20less
[2] files.catbox.moe/u2nvq1.pdf
[3] arxiv.org/pdf/0805.1400
+ In his 1900 work, Poincare pointed this out, so in a sense he also thought of m/E=1/c^2 back then, an equivalent relation.
* Poincare's proof is amazing though and deserves to be read. He formulated a model of a particle as constituent of charge construction, yet having zero internal field so it doesn't fly apart. Then he calculated the pressure required to stop it from being pulled apart by its own external field. Absolutely ingenious line of thinking, even though it is theory laden in that it assumes a structure of a particle.
** Einstein plagiarized Hilbert on the way out, but that's another priority dispute one that I'm not interested in (because GR is trash and Hilbert deserves it for helping contribute to that mess lol). The two had a bit of a love-hate relationship. Hilbert taunting Einstein, "every boy in the streets of Göttingen knows more about four-dimensional geometry than Einstein".

Einstein quipping back: "The people at Göttingen sometimes strike me, not as if they want to help one formulate something clearly, but as if they only want to show us physicists how much brighter they are than we".

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